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Regional Differentiation in Genetic Components for the American Beech, *Fagus grandifolia* Ehrh., in Relation to Geological History and Mode of Reproduction

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To demonstrate geographic patterns of genetic differentiation, 21 populations of the American beech, *Fagus grandifolia*, were investigated throughout its geographical range, using allozyme polymorphisms. A total of 1,131 trees from 21 populations were examined for 32 alleles of 10 polymorphic and two monomorphic loci in eight enzyme systems. The mean expected heterozygosity was 0.186, which indicates a relatively high genetic diversity within the populations. The levels of population differentiation were high, as revealed by genetic parameters, i.e., $G_{ST}=0.168$ and $F_{ST}=0.167$. The results of principal component analysis on allele frequencies clearly revealed unique regional patterns of differentiation in genetic components among populations "with" and "without" vegetative regeneration by root suckers. The American beech populations consist of two genetically distinct clusters, one from the Gulf-coastal plain, eastern coastal plain, Piedmont Plateau and Ozark Plateau; and the other from the remaining northern glaciated territories. Populations from the Blue Ridge and Great Smoky Mountains turned out to belong to the latter cluster, which is also characterized by extensive regeneration via root suckers. The consequences of regional differentiation in genetic components are discussed in relation to the post-glacial spread from refugia to the current geographic distributions and the mode of reproduction.

Key words: Allozyme — *Fagus grandifolia* — *F*-statistics — Geographic variation — Glacial refugia — Root sucker

A number of studies have been conducted on the genetic differentiation of local populations of particular woody species throughout their total geographical range, using allozymes as genetic markers (e.g. Hamrick and Godt 1989, 1997, Tomaru *et al.* 1997, Comps *et al.* 1990, Leonardi and Menozzi 1995, Konnert 1995, Belletti and Lanteri 1996, Hazler *et al.* 1997, Parks *et al.* 1994). However, recent progress in population analyses using different sampling strategies and molecular markers have brought new insights into the genetic differentiation within and among populations, as well as the occurrence of unique localized demographic genetic substructuring in relation to changing environmental constraints in habitats (Nei 1989, Hamrick and Nason 1996, Hanski and Gilpin 1997, Kitamura *et al.* 1997a, b, 1998, Kawano and Kitamura 1997). Woody species have long life-spans, often exceeding 300 years, and thus are ideal for investigating dynamic processes of spatio-temporal genetic changes in plant populations, because the effects of past and present reproductive events (roles of sexual and/or asexual reproduction; mast flowering and/or fruiting), and spatial as well as temporal demographic genetic substructurings within and among populations are well manifested (Ritland 1989, Alvarez-Buylla and Garay 1994, Alvarez-Buylla *et al.* 1996, Kitamura *et al.* 1997a, b, Kawano and Kitamura 1997, Kitamura *et al.* 2001 in press).

In connection with critical analyses on the demographic-genetic substructurings of local populations of American beech (*Fagus grandifolia*) in eastern North America, employing new approaches as argued above (Kitamura *et al.*, 1998, 2000 & 2001 in press), we too have focused on the genetic variations of local populations over the entire range, and in this paper document the genetic differentiation of this species in light of allozyme polymorphisms.

In this series of studies, we have selected the American beech for analyses of the demographic genetic substructuring of local populations and have also attempted to demonstrate patterns of population differentiation both on a small local scale (Kitamura *et al.* 1998, 2000) as well as over its extensive geographical range. The American beech (*Fagus grandifolia* Ehrh.) is one of the representative elements of eastern North American deciduous forests, and is the only species of the genus in North America (Fowells 1965). *Fagus* populations in the Mexican mountains facing the Gulf of Mexico were once regarded as an independent species, *Fagus mexicana* (Martinez 1940), or as an infraspecific variant of *F. grandifolia* (e.g. ssp. *mexicana*) (Little 1965), but are now included in *F. grandifolia* (Camp 1950, Shen 1992, Peters 1997).

Camp (1950) first distinguished three different races within the geographic range of the American beech. The races

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were named "gray", "red", and "white" beech. According to Camp (1950), the "gray beech" is a northern form found from Nova Scotia to the Great Lakes and on higher elevations of the Appalachian Mountains. The "white beech" is found on the southern Coastal Plain and foothills, extending northward along the Coastal Plain and Piedmont to about the glacial boundary. Between these two forms, mainly at mid-elevations on the southern Appalachian slopes, the "red beech" is common. However, race ranges overlap and occasionally co-occur within the population (Fowells 1965). Based upon the examination of material in North Carolina, Hardin and Johnson (1985) suggested that these variations may be broadly clinal but can only be demonstrated statistically, with much variation indicative of the other races within most populations. However, no evidence is available on the genetic background of differentiation among these races.

The American beech is monoecious (Brown 1922). The breeding system is an obligate xenogamy, since the outcrossing rate is estimated to be 1.0 (Kitamura *et al.* 1998). Seed dispersal is typical of a barochory and secondary dispersals are made by blue jays, *Cyanocitta cristata* (Johnson and Adkisson 1985), and rodents (Rohrig and Ulrich 1991). The American beech is shade-tolerant (Batista *et al.* 1998) and juveniles may survive more than 70 years of shade suppression (Poulson and Platt 1989, Canham 1990, also see Kitamura *et al.* 1997a). Mast flowering and fruiting is well known in beeches, with different intervals in different species: in the American beech at two- or three-year intervals (Deen 1936), and in the Siebold's beech (*Fagus crenata*) at three to seven year intervals (Kikuchi 1968, Peters 1997). Two fully ripe seeds are normally formed within each cupule in a regular mast fruiting year, but when flowering occurs in the succeeding season after regular mast fruiting, the levels of fecundity drop sharply, no embryos being formed in each flower, or each cupule producing only a single or no ripe seed, due to the lack of a sufficient energy supply (Isagi *et al.* 1997).

The American beech has two modes of regeneration (Ward 1961, Fowells 1965, Jones and Raynal 1986, 1988). One is sexual reproduction by seed and the other an asexual reproduction by root suckers. The occurrence of root suckers is obviously affected by microhabitat conditions in different geographic areas (Ward 1961). In general, populations of northern latitudes and higher elevations, where the environmental conditions are harsh, tend to produce root suckers (Held 1983, Kitamura et *al.* 2000). It is also evident that sexual reproduction is common in stable soil and shallow or exposed roots occasionally produce root suckers (Held 1983). Although abrasion of the roots may stimulate development of root sprouts (Ward 1961, Jones and Raynal 1988), root suckers are also often present where no injury is evident (Lutz 1930, Plice and Hedden 1931, Westveld 1933, Rushmore 1956).

Houston and Houston (1994) carried out a genetic study of the American beech populations in Massachusetts and West Virginia using nine polymorphic allozyme loci to reveal different patterns of clonal structures between the two populations in relation to the infection patterns of beech bark disease. Jones and Raynal (1986, 1988) also reported on the range of root suckering and completed a quantitative study on the two modes of reproduction, viz., the ratio of seed reproduction to root sucker regeneration. We have discovered that root suckering contributes to the maintenance of populations, and furthermore, plays a significant role in localized genetic substructuring, based upon critical mapping and size-class studies, not only of mature individuals, but also of juveniles at various growth stages and seedlings within study plots (Kitamura *et al.* 1998, 2000).

The geological succession of American beech populations during the post-Pleistocene glacial retreat has been demonstrated by pollen analyses (Davis 1983, Delcourt and Delcourt 1987). Whether or not the regeneration patterns of the American beech are related to postglacial successional processes over heavily glaciated as well as unglaciated territories is an important question, and thus the genetic background of regeneration and differentiation over the total geographical range needs to be critically investigated.

The questions being raised here are: i) what is the extent of genetic diversity in the American beech populations over the total geographic range?; ii) are the root suckering populations genetically different from other non-root suckering populations?; iii) is there any regional differentiation in genetic components as revealed by allozyme variability, reflecting the size and density of local beech populations and their background environments?

Materials and Methods

Study sites and sample collection

Twenty-one populations were chosen throughout the geographic range of the American beech (Table 1, Fig. 1). The total number of trees sampled was 1,131, which ranged from 20 to 117 trees per population.

We collected samples from May to August for five years from 1994 through 1998. The number of samples collected in each population is listed in Table 1. Random samplings were carried out except for QUE, MSH, MD, WG1, WG2, PDM, and GS, where mapping of all individuals within selected plots was conducted and a critical spatio-temporal analysis was carried out (Kitamura *et al.* 2000, 2001).

For samples from mapped populations, the following individual data were analyzed in this study. QUE was represented by mother trees whose diameters at breast height (dbh) were larger than 20 cm from two mapped transects on the same slope. In MSH, trees (dbh > 2.5 cm) in the mapped transect were selected and used for analysis. MD was represented by mother trees (dbh > 50 cm) within an area of 9 ha. WG1 was mapped with trees of dbh > 25 cm and WG2 with trees of dbh > 2.5 cm. GS represents mother trees of two mapped transects.

PDM represents a pooled data set of random samples at three sites where beech trees were scattered; Ivy Creek, Fluvanna, and along the Beech Grove Rd.

We collected several fresh leaves from each tree, brought them into the laboratory under low temperature conditions with ice, and stored them in deep freezer (−80 C) until

Table 1. Study sites for *Fagus grandifolia*

Code	Study area	N[1]	Group
QUE	Quebec City, Quebec	34	RS[2]
HD	Montreal, Quebec	20	RS
MSH	Mt. St. Hilaire, Quebec	74	RS
AQ	Algonquin Forest, Ontario	52	RS
OL	Oaster Lake, Ontario	52	RS
CR	Carolinean Forest, Ontario	52	RS
NH	Bridgetown, New Hampshire	52	RS
VT	Westbridgewater, Vermont	50	RS
PWD	Powdermill, Pennsylvania	46	RS
MD	Edgewater, Maryland	44	EC[3]
WG1	Wintergreen, Virginia	26	RS
WG2	Wintergreen, Virginia	34	EC
PDM	Piedmont, Virginia	50	EC
GS	Great Smoky Mts., Tennessee	20	RS
AC	Alum Cove, Arkansas	62	OP[4]
LMF	Little MO Falls, Arkansas	59	OP
OZ	Ozark Wilderness Area, Arkansas	104	OP
CC	Clark Creek, Mississippi	76	GC[5]
RH	Red Hill, Mississippi	55	GC
SMU	Miss. Univ. Property, Mississippi	117	GC
APL	Appalachicola, Florida	52	GC
Total		1,131	

[1] Number of samples, [2] population with root sucker, [3] population at East central Coastal Plain, [4] population at Ozark Plateau, [5] population at Gulf Coastal Plain.

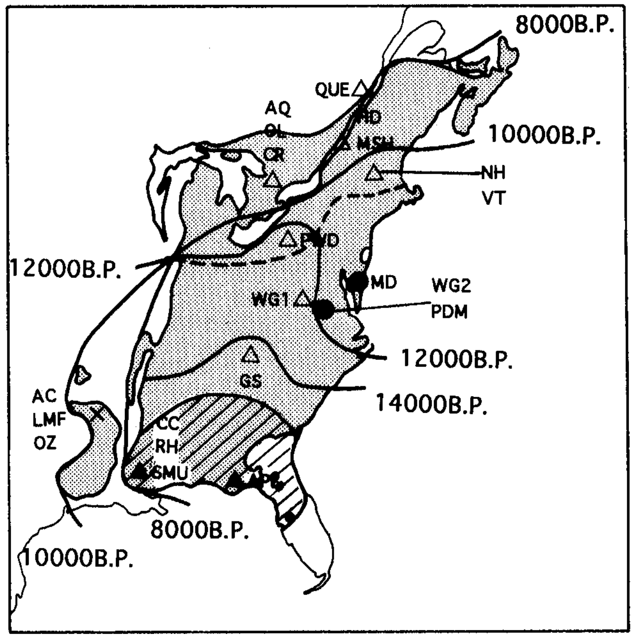

Fig. 1. Study sites and the extension of the North American beech after the last Pleistocene glaciation. Numbers indicate thousand of years B.P. Dotted line is the glacial boundary at 16000 B.P. (Delcourt and Delcourt 1987). Present distribution is shaded (Fowells 1965). Glacial refuge is hatched (Peters 1997). Open triangles, root sucker population; filled circles, East central coastal plain; filled triangles, Gulf coastal plain; crosses, Ozark plateau.

enzymes were extracted.

Enzyme extraction and electrophoreses

One hundred milligrams of leaf tissue was ground to a powder with liquid nitrogen and homogenized in one milliliter of extract buffer made up of 0.1 M Tris-HCl (pH 7.5), 25% (v/v) glycerol, 1% (v/v) Tween 80, 10 mM DTT, 1% (v/v) 2-mercaptoethanol and 70 mg/ml PVPP (Shiraishi 1988). The homogenates were centrifuged and 15 microlitres of the resulting supernatant was subjected to electrophoresis for each enzyme.

Polyacrylamide vertical slab gel electrophoresis was performed (Davis 1964, Orstein 1964) at 4 C, 12.3 mA/cm^2 for 150 minutes. The procedures used to detect the enzymes followed Richardson *et al.* (1986) and Shiraishi (1988). Putative loci were determined based on the quaternary structure of each enzyme (Weeden and Wendel 1989), and phenotypes were interpreted into genotypes.

Data analysis

Allele frequencies for each population were calculated from genotype frequencies. The expected heterozygosity (Nei and Royshoudhury 1974) and populational gene differentiation (G_{ST}; Nei 1973) were calculated based on the allele frequency. F-statistics (Wright 1922, 1965) and genetic distance and identity statistics (Nei 1972) were calculated using FSTAT by Goudet (1995) and GDID by Ritland (1989), respectively.

Principal component analysis (PCA) based on the correlation matrix and cluster analysis of population allele frequencies was done using SYSTAT ver. 5.2 (1992).

In order to clarify regional genetic differentiation, 21 sampled populations were categorized into four groups; i) population showing root suckers (QUE, HD, MSH, AQ, OL, CR, NH, VT, PWD, WG1, and GS); ii) Ozark Plateau (AC, LMF, and OZ); iii) Gulf Coastal Plain (CC, RH, SMU, and APL); iv) East Central Coastal Plain (MD, WG2, and PDM).

Results

Allozyme variations and their genotypic interpretation

Starch gel electrophoreses of the American beech revealed that nine loci for seven enzyme systems were polymorphic; phosphoglucose isomerase, peroxidase, phosphoglucomutase, alcohol dehydrogenase, malate dehydrogenase, and two loci each in 6-phosphogluconate dehydrogenase and cytochrome oxidase (Houston and Houston 1994).

We scored, based on polyacrylamide gel electrophoresis, 19 putative loci in 11 enzyme systems (Fig. 2); 6-phosphoglucose dehydrogenase (*6Pgdh1, 2, 3*), acid phosphatase (*Acp1, 2*), alcohol dehydrogenase (*Adh1, 2*), amylase (*Amy1, 2*), diaphorase (*Dia*), fumarase (*Fum*), glutamate oxaloacetate transaminase (*Got1, 2, 3*), leucine aminopeptidase (*Lap*),

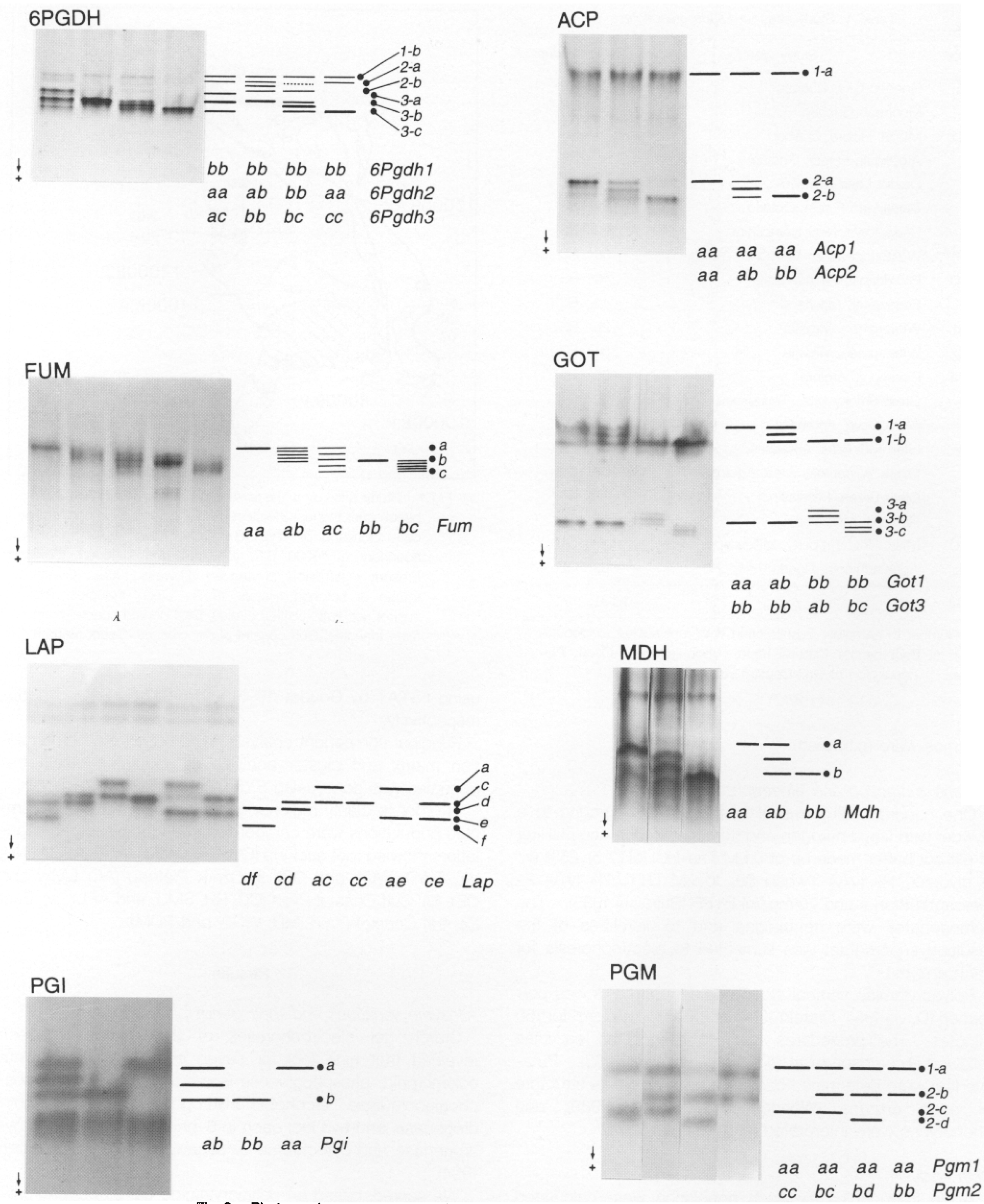

Fig. 2. Photographs, zymograms and genotypic interpretations for eight enzyme systems.

malate dehydrogenase (*Mdh*), phosphoglucose isomerase (*Pgi*), and phosphoglucomutase (*Pgm1, 2*), of which two (*Adh2* and *Pgm1*) were monomorphic and other three (*Acp1, Amy1*, and *Got2*) were unstable. A total of 41 alleles at 14 loci were detected throughout 21 populations of the American beech (Table 2). Due to poor enzyme activity, we failed to score *Got3* at WG1, *Mdh* at MD and GS, *Acp2* at MSH, HD, and NH, and *Amy2* at HD, NH, and MD; so that the remaining 32 alleles at 12 loci, including two monomorphic loci, were used for further analysis.

Genetic diversity and geographic differentiation of the American beech

Observations of allele frequencies indicate considerable differentiation among populations. Particularly noticeable are population differences in *6Pgdh2* (Fig. 3a), *6Pgdh3* (Fig. 3b), *Adh1, Fum, Lap* (Fig. 3c), and *Pgm2* (Fig. 3d) between populations "with" and "without" root suckers. This pattern is most clearly illustrated by the frequencies of *Pgm2-c* (Fig. 3d) and *6Pgdh2-b* (Fig. 3a).

Regional differences in less frequent or rare alleles are also observed especially in *6Pgdh1, Dia, Got1*, and *Pgi*. It is notable that less frequent alleles of *Got1* (*a* and *c*) are maintained in populations of the Gulf Coastal Plain and the Ozark Plateau. Rare alleles of *6Pgdh1* (*a* and *c*) and *Pgi* (*c* and *d*) are maintained in northernmost populations. Only the OZ population shows polymorphism in *Dia*.

The expected heterozygosity for each locus is shown in Table 3. Averaged across the populations, the mean expected heterozygosity (H_{exp}) is 0.186 (Table 3), indicating that considerable variation is maintained throughout the American beech populations. Regarding the populations with root suckers, the expected heterozygosities are high in *6Pgdh2, Lap*, and *Pgm2*, and low in *6Pgdh3, Fum* and *Adh1*, reflecting the regional differences in allele frequencies of these loci. A difference in the level of genetic diversity is not observed between populations with and without root suckers. Genetic differentiation among the 21 populations is high, with $G_{st}=0.168$.

Other measures for population differentiation were conducted by Wright's *F*-statistics. *F*-statistics for the total population over all loci are: $F_{IT}=0.146$, $F_{ST}=0.167$, and $F_{IS}=-0.026$ in which F_{IT} and F_{ST} are highly significant ($p<0.01$). The F_{ST} for all 10 loci show significant values (Table 4). These significant *F*-statistics indicate genetic differentiation of the populations in all of the allozyme loci used in this study. The inbreeding coefficient (F_{IS}) for each population does not show a significant deviation from zero, except for three populations (HD, OL, and NH) with root suckers (Table 4). Most of the positive significant F_{IS} values for each locus are observed in populations with root suckers (Table 4).

Principal component analysis and cluster analysis

Eigenvalues, proportions, and cumulative proportions for PCA of 30 allele frequencies are shown in Table 5. The proportions of the Z_1 and Z_2 components were low, 25.32% and 13.33%, respectively. Figure 4 shows the factor loadings for Z_1 and Z_2 of PCA. High positive factor loadings were attained for Z_1 in *Lap-a, Pgm2-c, 6Pgdh2-b, Fum-b* and *Adh1-b*, which are characteristic of root sucker populations. On the other hand, negative factors for Z_1 were observed in *6Pgdh2-a, Pgm2-b, Lap-e, Fum-a, Adh1-a, 6Pgdh3-c*, and *Pgi-b*, which are characteristic alleles for populations without root suckers. The high positive factor loadings for Z_2 are due to the presence of specific alleles in the AQ (*Lap-c, 6Pgdh1-a*, and *Pgi-e*) and OZ (*6Pgdh2-a*) populations.

The projection diagram for Z_1 and Z_2 components is shown in Fig. 5. Although the cumulative proportions are low (Table 5), the projection diagram clearly indicates the difference between populations "with" and "without" root suckers by Z_1 component. Populations with root suckers tend to have positive Z_1 component scores, while populations without root suckers have negative values.

We calculated Nei's genetic identity and distance to determine the genetic relationships among populations (Table 6). The clusterings of 21 populations are shown in Fig. 6. The dendrogram clearly shows the two major groups of populations which relate to whether root suckers are extensive or not: i. e., populations which occur in the East-Central Coastal Plain, Gulf Coastal Plain and Ozark Plateau, and those which occur primarily in the heavily glaciated northern territories and also on the mountain ridges exposed to severe climatic changes during and/or after the Pleistocene Ice Age (Delcourt and Delcourt 1987).

Discussion

Genetic diversity of the American beech

One of the advantages of selecting a representative dominant tree species of North America is the availability of much basic information concerning exact range distribution, phytosociology, reproductive systems as well as past geological succession. This ensures that the new data obtained can be examined in the light of the results of earlier studies (Camp 1950, Shelford 1963, Fowells 1965, Forcier 1975, Hardin and Johnson 1985, Davis 1983, Delcourt and Delcourt 1987).

The levels of genetic diversity within and/or among populations of a given species are influenced by various factors, including breeding systems, asexual reproductive systems, and various other life history and demographic features (Kawano 1975, 1985, Harper 1977, Hamrick and Godt 1989). The environmental regimes of the habitats as well as selective forces are also responsible for the fixation of genetic diversities of local plant populations. As noted above, the American beech is one of the representative climax elements of the cool- to warm-temperate forests in eastern North America, sharing habitats with various deciduous or evergreen broad-leaved as well as coniferous trees (Fowells 1965, Forcier 1975, Rohrig and Ulrich 1991, Russell 1953, Shelford 1963).

Beeches have long life-spans, often extending well over 300 years, characterized by unique masting in seed production, and thus irregularly formed gaps within the forest stands are known to provide significant opportunities for new generations within the habitats (Russell 1953, Sain and Blum 1981,

Table 2. Allele frequencies of 14 loci for 21 populations

Locus	Allele	QUE	HD	MSH	AQ	OL	CR	NH	VT	PWD	MD	WG1	WG2	PDM	GS	AC	LMF	OZ	CC	RH	SMU	APL	Total
Lap	*a*	.324	.225	.014	.135	.279	.029	.106	.130	.000	.125	.462	.000	.010	.325	.000	.000	.000	.000	.000	.000	.000	.067
	b	.000	.000	.000	.000	.000	.000	.000	.050	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.002
	c	.515	.600	.304	.798	.481	.452	.394	.430	.344	.341	.481	.515	.550	.350	.540	.576	.611	.539	.664	.491	.558	.508
	d	.029	.025	.034	.000	.048	.106	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.011
	e	.132	.150	.507	.067	.154	.413	.490	.380	.489	.534	.058	.485	.440	.325	.460	.424	.389	.461	.336	.509	.442	.393
	f	.000	.000	.142	.000	.038	.000	.010	.010	.167	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.019
6Pgdh1	*a*	.000	.000	.000	.010	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	b	1.000	1.000	.847	.990	.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.990	.990
	c	.000	.000	.153	.000	.010	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.010	.010
6Pgdh2	*a*	.147	.200	.371	.596	.269	.000	.206	.212	.438	.568	.457	.603	.531	.350	.903	.576	.944	.664	.583	.705	.644	.541
	b	.853	.800	.629	.404	.731	1.000	.794	.788	.563	.432	.543	.397	.469	.650	.097	.424	.056	.336	.417	.295	.356	.459
6Pgdh3	*a*	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.007	.000	.000	.000	.000
	b	.971	.950	.993	1.000	.962	.933	.731	.840	.989	.557	.938	.735	.870	.925	.935	.466	.942	.362	.382	.568	.394	.756
	c	.029	.050	.007	.000	.038	.067	.269	.160	.011	.443	.063	.265	.130	.075	.065	.534	.058	.632	.618	.432	.606	.243
Fum	*a*	.029	.100	.400	.000	.096	.038	.048	.180	.075	.091	.000	.397	.418	.000	.403	.537	.115	.253	.173	.167	.163	.190
	b	.897	.900	.580	.750	.856	.933	.923	.740	.850	.875	1.000	.485	.531	1.000	.556	.398	.832	.687	.764	.722	.625	.737
	c	.074	.000	.020	.250	.048	.029	.029	.080	.075	.034	.000	.118	.051	.000	.040	.065	.053	.060	.064	.111	.212	.074
Pgm	*b*	.779	.700	.932	.885	.731	.904	.865	.670	1.000	.966	1.000	.926	.970	.825	1.000	1.000	1.000	.967	.991	.957	1.000	.924
	c	.221	.300	.068	.115	.269	.096	.135	.330	.000	.034	.000	.074	.030	.175	.000	.000	.000	.033	.009	.043	.000	.076
Pgi	*b*	.853	.975	.864	.808	.962	1.000	.962	.960	1.000	1.000	.769	.912	1.000	1.000	.992	1.000	.990	1.000	.855	.987	1.000	.954
	c	.015	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000
	d	.044	.000	.007	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.002
	e	.088	.025	.129	.192	.038	.000	.038	.040	.000	.000	.231	.088	.000	.000	.008	.000	.010	.000	.145	.013	.000	.043
Adh1	*a*	.015	.000	.000	.000	.000	.038	.010	.050	.050	.034	.000	.015	.224	.000	.024	.276	.038	.072	.027	.094	.019	.054
	b	.985	1.000	1.000	1.000	1.000	.962	.990	.950	.950	.966	1.000	.985	.776	1.000	.976	.724	.962	.928	.973	.906	.981	.946
Got1	*a*	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.016	.000	.019	.000	.000	.026	.000	.005
	b	1.000	1.000	1.000	1.000	1.000	.981	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.984	.983	.981	1.000	1.000	.974	1.000	.993
	c	.000	.000	.000	.000	.000	.019	.000	.000	.000	.000	.000	.000	.000	.000	.000	.017	.000	.000	.000	.000	.000	.002
Got3	*a*	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000		.000	.020	.000	.000	.000	.000	.013	.000	.017	.000	.004
	b	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000		.985	.980	1.000	1.000	1.000	1.000	.987	1.000	.970	.971	.993
	c	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000		.015	.000	.000	.000	.000	.000	.000	.000	.013	.029	.003
Mdh	*a*	.162	.325	.047	.317	.144	.048	.030	.180	.000		.000	.000	.000		.175	.036	.202	.086	.109	.239	.000	.122
	b	.838	.675	.953	.683	.856	.952	.950	.820	1.000		1.000	1.000	1.000		.775	.696	.755	.789	.882	.761	1.000	.845
	c	.000	.000	.000	.000	.000	.000	.020	.000	.000		.000	.000	.000		.050	.268	.043	.125	.009	.000	.000	.032
Dia	*b*	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.865	1.000	1.000	1.000	1.000	.987
	c	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.135	.000	.000	.000	.000	.013
Acp2	*a*	.875			.865	.806	.904		1.000	.725	.807	1.000	.853	.980	.825	.992	.932	.918	.586	.900	.803	.875	.865
	b	.125			.135	.173	.077		.000	.275	.193	.000	.147	.020	.175	.008	.068	.082	.414	.100	.197	.125	.133
	c	.000			.000	.020	.019		.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.002
Amy2	*a*	.931		.973	1.000	.990	1.000		1.000	1.000		1.000	1.000	.910	.900	.992	.975	.971	.875	.982	.902	.981	.960
	b	.069		.027	.000	.010	.000		.000	.000		.000	.000	.090	.100	.008	.025	.029	.125	.018	.098	.019	.040

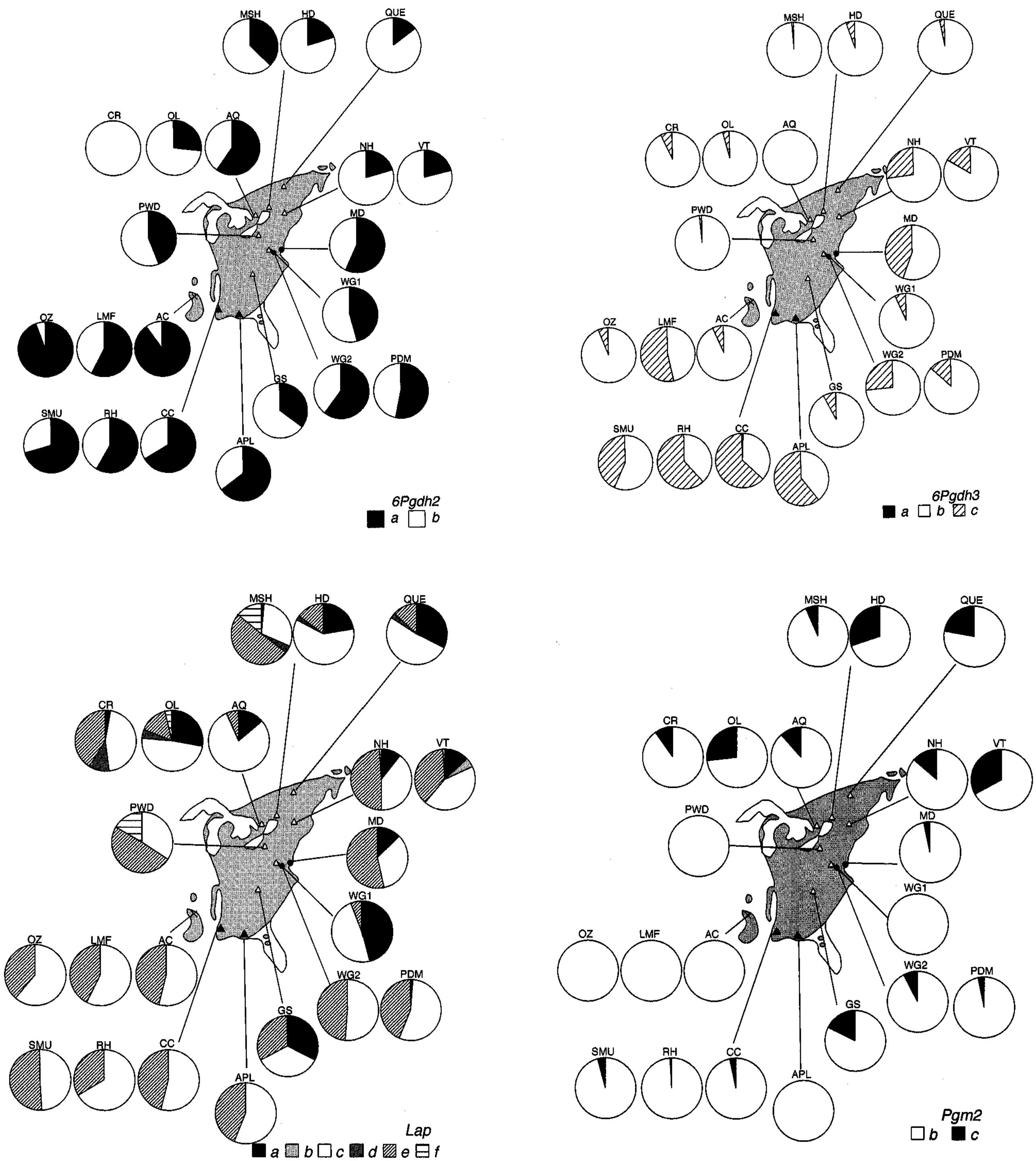

Fig. 3. Allele frequencies for 21 populations. a) *6Pgdh2*; b) *6Pgdh3*; c) *Lap*; d) *Pgm2*.

Held 1983, Nielsen and Schaffakitzky de Muckadeli 1954, Nakashizuka 1987, 1988, Yamamoto 1989, Peters 1997, Kawano and Kitamura 1997). The historical changes in range expansion of the American beech populations during the past 20,000 years have been demonstrated by Delcourt and Delcourt (1987) using extensive pollen analysis. In the light of all this information including ecological features of habitats, life history parameters, and past range expansions (or contractions) and migration during the geological period, we can understand the levels of genetic diversity found in contemporary beech populations over their total geographical range in relation to the ecological background of population differentiations.

The results of the present study have revealed that

Table 3. Genetic differentiation (F_{ST}) between study sites

Between study sites	F_{ST}	
WG1 & WG2	0.460	($p<0.01$)
GS1 & GS2	0.234	($p<0.01$)
WG1 & GS1	0.354	($p<0.01$)

American beech populations comprise a considerable amount of genetic variation throughout its geographic range (Tables 3 and 4). The mean expected heterozygosity within populations for American beech was 0.186 (Table 3), which is much higher than the average value for long-lived woody plants (0.149) reviewed by Hamrick and Godt (1989). Among *Fagus* species, the heterozygosity of the American beech is as high as that of Siebold's beech, *F. crenata* (0.187; Tomaru *et al.* 1997), and within the range of that of the European beech, *F. sylvatica* (0.136-0.308) (Comps *et al.* 1990, Leonardi and Menozzi 1995, Konnert 1995, Belletti and Lanteri 1996, Hazler *et al.* 1997), but much lower than that of *F. multinervis* endemic to Ullung Island, Korea (0.230 in Chung *et al.* 1998, 0.373 in Ohkawa *et al.*, unpubl. data).

Genetic differentiation is measured by parameters such as G_{ST} and F_{ST}. A review by Hamrick and Godt (1989) on various plant populations indicated that $G_{ST}=0.076$ for long-lived woody plants. The G_{ST} value for Siebold's beech over the total range in the Japanese Islands is 0.038 (Tomaru *et al.* 1997), but being 0.016 for large montane populations and 0.107 for localized small lowland populations in Toyama Prefecture, central Honshu, Japan (Ohkawa *et al.* 1998). More or less similar G_{ST} values are known for the European beech, i.e., 0.043 (Belletti and Lanteri 1996) and 0.019 (Konnert 1995). Also the F_{ST} values for the European beech are 0.014-0.054 (Comps *et al.* 1990, Leonardi and Menozzi 1995, Hazler *et al.* 1997). By comparison, therefore, the differentiation among the American beech populations is very high with $G_{ST}=0.168$ and $F_{ST}=0.167$ (Table 4). However, an unusually high G_{ST} value, 0.963, was obtained by mtDNA analysis for Siebold's beech populations, and was interpreted to indicate extensive geographical populational differentiation (Tomaru *et al.* 1998).

A high G_{ST} value (0.168) obtained for American beech may reflect the differences in reproductive systems of local populations, i.e., effective vegetative reproduction by root sucker formation in addition to sexual reproduction, consequences of past migratory history during and/or after glacial expansion and retreat (Delcourt and Delcourt 1987), as well as the ecological backgrounds of the present-day local populations (Rohrig and Ulrich 1991). The sampled populations in this study are more or less from the entire geographic range of the American beech. Fixed genotypes of localized and isolated populations occasionally tend to result in higher values of G_{ST}, reflecting genetic drift or founder effect (Crow and Kimura 1970). In this respect, it is also possible that G_{ST} values reflect past successional as well as ecological backgrounds of local populations in different regions of the North American continent.

Houston and Houston (1994) analyzed the genetic structure of two northern populations (West Virginia and Massachusetts) of the American beech, using nine allozyme loci, and revealed that the F_{ST} between these two populations was very low, 0.064, obviously reflecting the predominant recruitment of offspring by root sucker formation. Their results may also reflect that the two populations sampled are in close proximity, and thus there may be no sharp genetic differentiation between them. The present results on the 21 different populations from the overall geographic range, however, demonstrated a much higher F_{ST} value of 0.167, reflecting genetic subdivisions among populations.

One of the reasons for the high F_{ST} value found in American beech is attributed to regional allelic differences, reflecting the differences in allele frequencies among populations (e.g., Fig. 3). The northern populations maintain rare alleles of *6Pgdh1* and *Pgi*, and also have high polymorphisms in *Lap*. Two populations from the east Coastal Plain (WG2 and PDM) maintain rare alleles of *Got3*. Gulf Coastal Plain populations maintain less frequent alleles of *Got1*, *Got3*, and *6Pgdh1*. Populations with a fragmented distribution on the Ozark Plateau have polymorphisms in *Dia* and rare alleles of *Got1*.

Comparisons of allelic components suggested differences between northern and southern populations at several loci. For northern populations, allele frequencies were high at *Pgm2-c*, *6Pgdh2-b*, *Lap-a*, and *Adh1-a*, but low at *6Pgdh3-c* and *Fum-a* (Table 2). It is also notable that the two populations from the Blue Ridge (WG1) and Great Smoky Mts. (GS), tended to have the same allelic components as northern populations (Table 2, Fig. 3). These two populations showed extensive root sucker formation which is also characteristic of the northern populations (Kitamura *et al.* 2001 in press). This evidence suggests that the sharp differences detected between the northern-montane ('with' root sucker formation) and the southern-Piedmont-coastal ('without' root suckers) populations in the allele frequencies may reflect an extended period of isolation, i.e., genetic variations that differentiated at earlier stages during range extension are still conserved in these populations.

Earlier studies reported evidence of allozyme-habitat associations (reviewed by Hoffman *et al.* 1995, Lonn 1993, Lonn *et al.* 1996); notably unique parapatric genetic differentiation related to edaphic conditions (Nevo *et al.* 1994), such as soil moisture (Zangerl and Bazzaz 1984), soil nutrients (Xie and Knowles 1992), and calcareous rock (Heywood and Levin 1985) have been reported. Our recent analyses of the allozyme polymorphism of *Poa annua*, a cosmopolitan annual weed, also revealed the effects of strong selection pressure on paddy field populations in Japan, representing a unique example of parapatric differentiation in neighboring habitats (Kawano unpubl., Akimoto and Kawano unpubl. and in prep.). Although the presence of point mutations in a single gene with strong adaptive roles has been reported for a number of organisms (Nei 1989), including *Poa annua* (Barros and Dyer 1988), the functional aspects of the above-mentioned allozyme genes have not yet been clarified in relation to a specific selection regime.

In this study, we have questioned whether or not there is

Table 4. Inbreeding coefficient (F_{IS}) and fixation index (F_{ST}) for 21 populations

Locus	F_{IS} per population											
	QUE	HD	MSH	AQ	OL	CR	NH	VT	PWD	MD	WG1	WG2
Lap	.150	.404*	−.002	.387**	.285**	−.246	.488**	.333**	−.584	−.237	.527**	−.580
6Pgdh1	—	—	−.173	.000	.000	—	—	—	—	—	—	—
6Pgdh2	.077	.701*	−.375	−.269	.325*	—	−.010	.212	−.765	.085	.058	.093
6Pgdh3	−.015	−.027	.000	—	−.030	−.063	−.360	−.181	.000	−.232	−.045	.108
Fum	−.074	−.086	.096	−.325	−.117	.257	.212	−.248	.273	−.100	—	−.326
Pgm2	−.269	.073	.575**	−.121	.032	.564**	.348*	.060	—	−.024	—	−.065
Pgi	−.106	.000	.203	−.229	−.030	—	−.030	.487	—	—	−.282	−.082
Adh1	.000	—	—	—	—	−.030	.000	−.043	−.027	−.024	—	.000
Got1	—	—	—	—	—	−.010	—	—	—	—	—	—
Dia	—	—	—	—	—	—	—	—	—	—	—	—
Over all loci	−.014	.288**	−.016	−.124	.158**	−.026	.144*	.077	−.473	−.123	.148	−.182

Table 4. Continued

Locus	F_{IS} per population									All	*Fst*
	PDM	GS	AC	LMF	OZ	CC	RH	SMU	APL		
Lap	−.062	−.481	−.063	−.172	−.026	.053	−.172	.098	−.004	.009	.076**
6Pgdh1	—	—	—	—	—	—	—	—	.000	−.149	.128**
6Pgdh2	.109	1.000**	.269	−.102	.522**	−.085	−.248	−.003	.045	.008	.263**
6Pgdh3	−.140	−.056	−.061	.157	.120	−.057	.008	−.093	−.158	−.072	.299**
Fum	−.311		−.218	−.037	−.049	.052	−.178	.010	−.063	−.089	.120**
Pgm2	−.021	−.188	—	—	—	.385	.000	−.040	—	.093**	.126**
Pgi	—	—	.000	—	−.005	—	−.015	−.009	—	−.032	.091**
Adh1	−.280	—	−.017	−.287	−.035	−.071	−.019	.001	−.010	−.125	.097**
Got1	—	—	−.008	−.009	−.015	—	—	.320	—	.116	.006*
Dia	—	—	—	—	−.068	—	—	—	—	−.063	.122**
Over all loci	−.126	.057	−.076	−.078	.015	.000	−.126	.009	−.045	−.026	.167**

any regional differentiation in genetic components as revealed by allozyme variability, reflecting the size and density of local beech populations and their background environments. The allozyme loci we have examined in the present studies may be either selectively neutral (e.g. Nei 1972) or subjected to natural selection (Nevo *et al.* 1979, 1982, 1986ab, Kitamura *et al.* 1997ab, Kawano and Kitamura 1997), but the preponderance of specific genotypes with root suckers suggests the effects of selection in harsher environments in glaciated territories as well as fragmented remnant habitats in the southern montane zones (e.g., the Great Smoky Mountains, and Blue Ridges) (Held 1983). The root sucker populations, which have been called "gray beech" (a northern race) (Camp 1950), nowadays have their geographical range throughout the heavily glaciated northern territories with a continuous narrow band into the Blue Ridge, Allegheny, and Great Smoky Mountains. It can be presumed that the range extensions of the American beech, which propagate vegetatively and invade surrounding locations most efficiently by root suckering, into northern glaciated territories have taken place repeatedly during post-glacial periods (Delcourt and Delcourt 1987). Differences in genetic composition between the populations "with" and "without" root suckers are clear among existing American beech populations from the results of the PC and cluster analyses (Figs. 5, 6).

We have been studying the fine-scale genetic structure of the local populations of Siebold's as well as American beech (Kitamura *et al.* 1997a, b, Kawano and Kitamura 1997, Kitamura *et al.* 1998, 2000). In this series of studies, we have been placing emphasis on analyzing demographic genetic differentiation within and/or among populations, and further interpreting the mechanisms of localized differentiation within patch or local populations. In this paper, we report the overall genetic diversity of the North American populations over the total geographic range based on allozyme polymorphisms of 21 populations. Our studies on the fine-scale genetic structures of local populations of the North American beech (Kitamura *et al.* 2000, Kitamura *et al.* 2001) clearly revealed extraordinary unique clonal patterns among root sucker populations, with clonal patchiness of various sizes. It is evident that root sucker formation is responsible for the unique spatio-temporal genetic substructures within local populations. If the populations were to

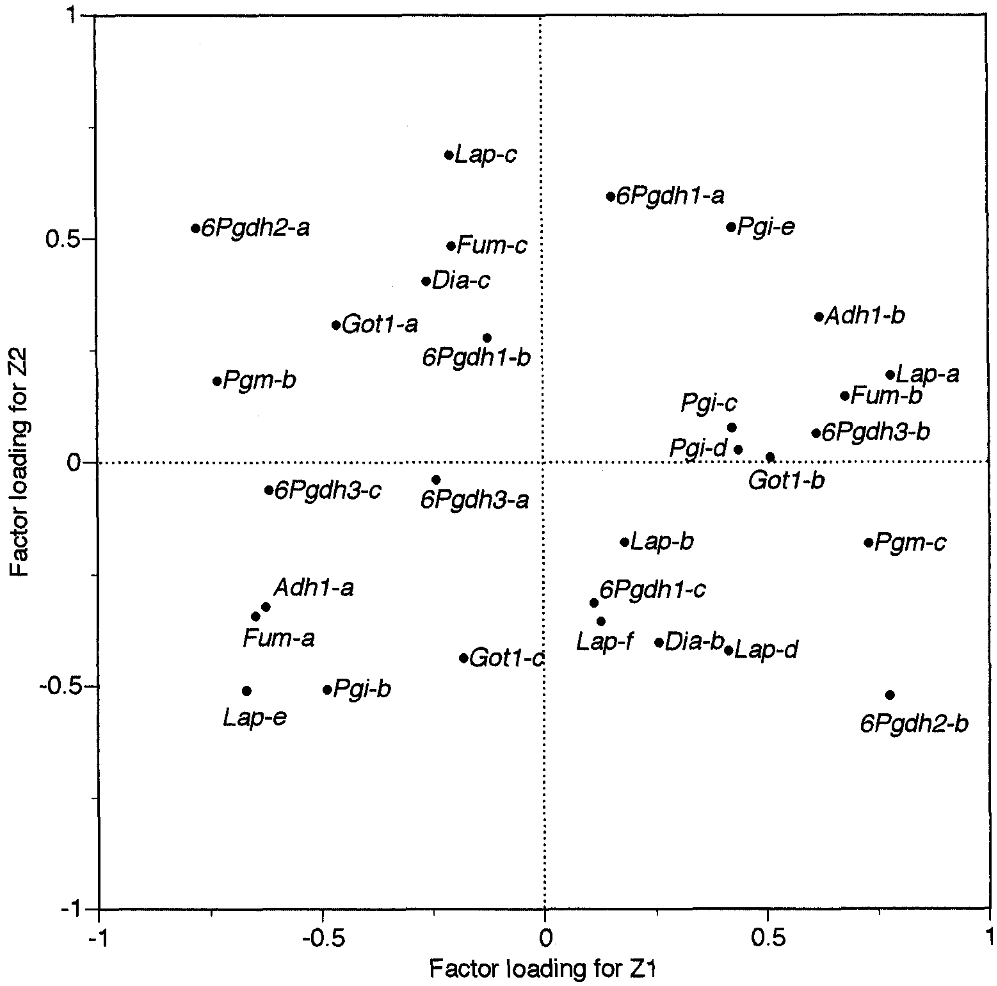

Fig. 4. Factor loadings for Z_1 and Z_2 of PCA (cf. Fig. 5) on 30 alleles of ten polymorphic loci for 21 populations.

regenerate only by root suckers, the levels of genetic variation may drop lower than those obtained without root suckers. Our results, however, did not reveal differences in the levels of the genetic variation among populations. This fact indicates the populations with root suckers regenerate sexually as well as asexually, and maintain the genetic variation within the population. The multilocus analysis of clonal structures clearly shows various kinds of clones are established within populations (Kitamura *et al.* 2000), suggesting that root suckering populations are not genetically uniform, and comprise a high level of genetic diversity. Similar genetic substructurings of local populations are reported in woodland clonal herbs (pseudo-annuals), *Disporum smilacinum* (Kawano, 1984) and *Uvularia perfoliata* (Kudoh *et al.* 1999), although on a much finer spatial scale within a local population.

Another remarkable finding for root suckering populations in northern territories is positive deviations in F_{IS} values (Table 4). F_{IS} is essentially a genetic parameter that indicates the level of inbreeding (Wright 1922, 1965), but if asexual (or vegetative) reproduction is predominant, and also population subdivisions due to clonal patchiness occur frequently, the F_{IS} values may deviate from zero as well, reflecting high levels of duplication of the same genotypes within a local population and violating the Hardy-Weinberg equilibrium. Positive significant values of F_{IS} over all loci were observed in HD, OL and NH populations which are northern root sucker populations. All of the positively significant values of F_{IS} for each locus were observed in root sucker populations except for *6Pgdh2* in OZ. However, our latest analysis for the Great Smoky populations demonstrated negatively significant F_{IS} values, suggesting an excess of heterozygotes (Kitamura *et al.* 2001).

Remarkable clonal distributions by extensive root sucker formation were found in QUE, MSH, PWD, WG1, and GS (Table 1). One of the most characteristic edaphic conditions of the glaciated territories is that the soil layers are very shallow with a well aerated rocky substratum, and thus root systems tend to extend in all possible directions, often being exposed at the ground surface. If certain genotypes are adapted to such specific micro-environments and are also able to propagate by root sucker, particular genotypes may become predominant within a patch population, resulting in high genetic heterogeneity and/or patchiness. The devia-

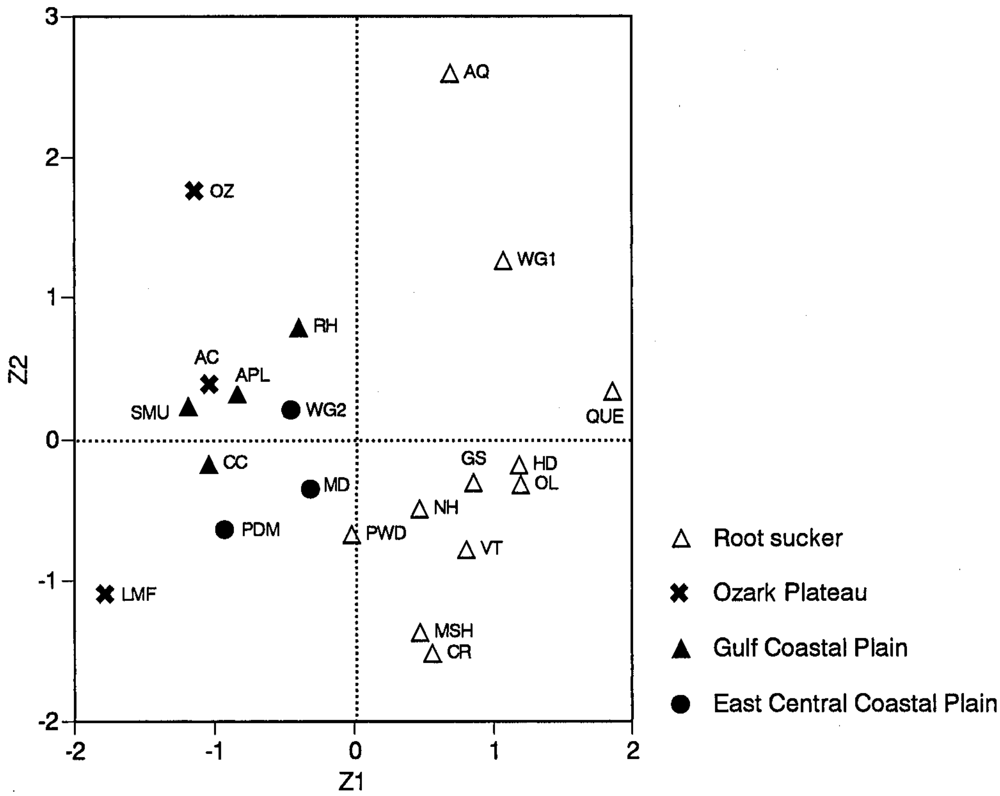

Fig. 5. Principal component analysis on 30 alleles of ten polymorphic loci for 21 populations.

Table 5. Eigenvalues, proportions, and cumulative proportions for PCA

	Eigenvalue	Proportions (%)	Cumulative proportions (%)
Z1	7.60	25.32	25.32
Z2	4.00	13.33	38.65

tions from Hardy-Weinberg equilibrium are indirect evidence for these population subdivisions due to clonal structures.

Population differentiations in Pleistocene refugial history

As revealed by the PCA, there are several alleles which are specific to the regional groups of populations. The uniqueness of the AQ population in Ontario may be attributed to the isolation of a small population within a pine plantation which then may have re-expanded. The unique genetic structure may be partly due to genetic drift, which is clearly shown by the PC analysis where only AQ is entirely isolated from the other root sucker populations (Fig. 5).

The overall geographic patterns of genetic diversity found in 21 regional populations obviously reflect the Pleistocene refugial history. Our results are consistent with the distributional history of the American beech interpreted by pollen analysis (Delcourt and Delcourt 1987).

The Gulf Coastal Plain has been regarded as one of the refugia for deciduous trees during the Pleistocene glaciation period (Parks *et al*. 1994). Peters (1997) also assumed that refugia for the American beech were located on the Gulf Coastal Plain. Appalachian elements were forced to migrate southward to the Gulf coastal areas during the glacial maxima (Davis 1983). Species that were characteristic of the Appalachian mountains have persisted on the cool bluffs of the Apalachicola River as isolated remnant populations from when these bluffs served as a refugium during Pleistocene glacial advances. The presence of these upland populations with other Appalachian plants on these coastal plain river bluffs is taken as additional evidence that these areas served as Pleistocene refugia for the Appalachian elements (Parks *et al*. 1994).

Fig. 1 demonstrates the presumed chronological changes in the distribution of the North American beech in the post -glacial period based on all the evidence available (Davis 1983, Delcourt and Delcourt 1987, Parks *et al*. 1994) and the geographical range of root sucker populations confirmed in the present study. The results of the cluster analysis suggest population extension of the American beech from the Gulf Coastal Plains to East-Central Coastal Plains. According to the pollen analysis data, the populations on the Ozark Plateau may have once had a continuous range with Coastal Plain populations (Delcourt and Delcourt 1987), and the results of the present study also indicate such a possibility. The somewhat discontinuous genetic components in Ozark Plateau populations (Fig. 6) suggest effects of genetic isolation due to subsequent fragmentation of populations. According to the fossil records, the American beech (*Fagus grandifolia*) once occurred on the west coast, and became

Table 6. Nei's genetic distance (upper triangle) and identities (lower triangle)

Population	QUE	HD	MSH	AQ	OL	CR	NH	VT	PWD	MD	WG1	WG2	PDM	GS	AC	LMF	OZ	CC	RH	SMU	APL
QUE		0.003	0.037	0.031	0.003	0.013	0.017	0.012	0.029	0.052	0.018	0.057	0.050	0.010	0.087	0.097	0.081	0.087	0.071	0.069	0.083
HD	0.997		0.036	0.029	0.002	0.014	0.016	0.008	0.028	0.048	0.024	0.049	0.043	0.010	0.076	0.086	0.071	0.076	0.064	0.061	0.074
MSH	0.963	0.965		0.044	0.029	0.032	0.027	0.023	0.014	0.039	0.044	0.020	0.018	0.030	0.036	0.055	0.053	0.061	0.061	0.043	0.059
AQ	0.970	0.971	0.957		0.025	0.055	0.046	0.040	0.031	0.049	0.020	0.034	0.036	0.031	0.038	0.076	0.029	0.064	0.049	0.043	0.056
OL	0.997	0.998	0.972	0.975		0.017	0.015	0.007	0.021	0.041	0.018	0.043	0.038	0.006	0.065	0.082	0.062	0.071	0.061	0.054	0.068
CR	0.987	0.986	0.968	0.946	0.984		0.009	0.013	0.022	0.048	0.042	0.058	0.047	0.018	0.095	0.090	0.092	0.083	0.072	0.067	0.079
NH	0.983	0.984	0.973	0.955	0.985	0.991		0.008	0.016	0.018	0.032	0.034	0.035	0.010	0.069	0.060	0.066	0.043	0.037	0.034	0.042
VT	0.988	0.992	0.977	0.961	0.993	0.987	0.992		0.022	0.033	0.038	0.032	0.030	0.013	0.066	0.062	0.071	0.057	0.052	0.044	0.056
PWD	0.971	0.973	0.986	0.970	0.979	0.978	0.984	0.978		0.022	0.029	0.025	0.019	0.014	0.034	0.060	0.032	0.050	0.049	0.029	0.047
MD	0.949	0.954	0.962	0.952	0.960	0.953	0.982	0.967	0.978		0.040	0.020	0.029	0.025	0.038	0.033	0.036	0.011	0.014	0.006	0.012
WG1	0.982	0.976	0.957	0.980	0.982	0.959	0.969	0.963	0.972	0.961		0.051	0.049	0.014	0.061	0.090	0.048	0.071	0.054	0.052	0.068
WG2	0.945	0.952	0.981	0.966	0.958	0.944	0.966	0.968	0.976	0.981	0.951		0.008	0.041	0.014	0.018	0.029	0.019	0.023	0.011	0.018
PDM	0.951	0.958	0.982	0.965	0.963	0.954	0.966	0.970	0.981	0.972	0.952	0.992		0.038	0.018	0.019	0.031	0.033	0.038	0.020	0.034
GS	0.990	0.990	0.971	0.969	0.994	0.982	0.990	0.988	0.986	0.975	0.986	0.959	0.963		0.059	0.080	0.050	0.060	0.055	0.042	0.059
AC	0.916	0.927	0.964	0.963	0.937	0.909	0.933	0.936	0.967	0.963	0.941	0.986	0.983	0.943		0.041	0.010	0.040	0.049	0.022	0.040
LMF	0.907	0.917	0.946	0.927	0.922	0.914	0.942	0.940	0.942	0.967	0.914	0.982	0.981	0.923	0.960		0.062	0.015	0.024	0.020	0.019
OZ	0.922	0.931	0.949	0.972	0.940	0.912	0.936	0.932	0.969	0.965	0.953	0.971	0.969	0.951	0.990	0.940		0.045	0.048	0.023	0.044
CC	0.917	0.927	0.941	0.938	0.931	0.921	0.958	0.945	0.951	0.989	0.931	0.981	0.967	0.941	0.961	0.985	0.956		0.005	0.005	0.002
RH	0.932	0.938	0.941	0.952	0.941	0.931	0.964	0.949	0.952	0.986	0.947	0.978	0.962	0.947	0.953	0.976	0.953	0.995		0.011	0.006
SMU	0.933	0.941	0.958	0.958	0.948	0.935	0.967	0.957	0.972	0.994	0.949	0.989	0.980	0.959	0.978	0.980	0.977	0.995	0.989		0.006
APL	0.921	0.929	0.942	0.946	0.934	0.924	0.959	0.946	0.954	0.988	0.934	0.982	0.966	0.943	0.961	0.981	0.957	0.998	0.994	0.994	

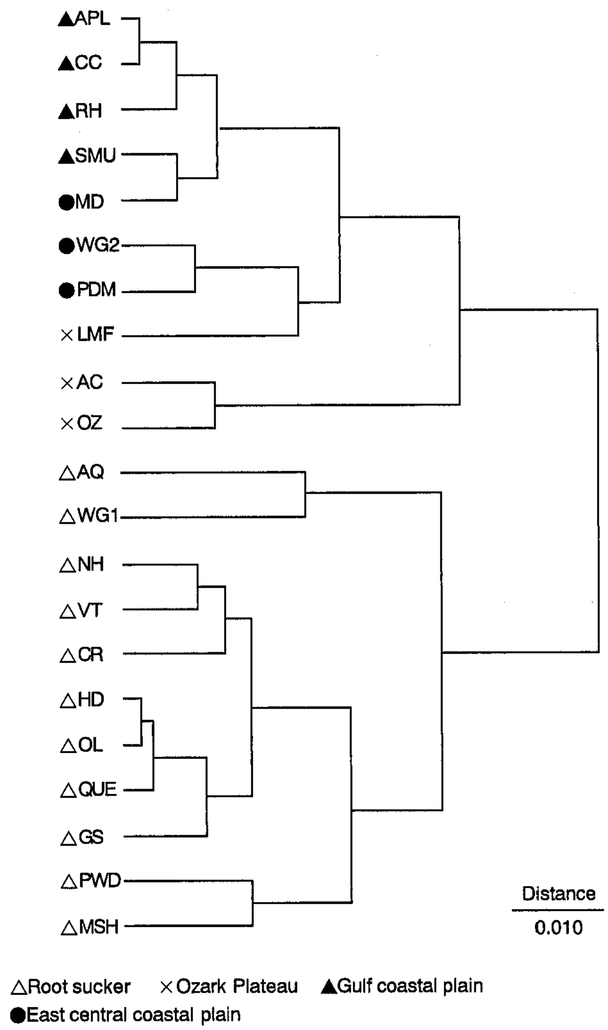

Fig. 6. Cluster analysis on 30 alleles of ten polymorphic loci for 21 populations.

extinct there before the glacial period (Chaney 1940, Robichaux and Taylor 1977), but there is no direct evidence of a genetic connection with contemporary Ozark or any other eastern populations.

According to pollen records (Delcourt and Delcourt 1987), American beech populations have extended their ranges into the northern Applachian Highlands and Interior Plateau ca. 16,000 to 12,000 BP, and further range extensions to the central Great Lakes region and Nova Scotia occurred 8,000 BP. Simultaneously, fragmentation and isolation of Ozark populations obviously occurred. Therefore, it is most probable that root sucker populations which have an adaptive advantage in growing on shallow and rocky soil layers, rapidly spread into the territories left after glacial retreat. The American beech primarily occupies moister, but mesic sites, whereas sugar maple and birches grow on somewhat drier sites, forming patchy mosaic communities (Kitamura and Kawano unpubl. obs.). Such unique local habitat segregation among woody elements of the northern mixed forests which have developed throughout the glaciated territories suggests that the difference in ecophysiological requirements (e.g. root susceptibility to soil moisture and nutrient levels, and aeration) of these species has resulted in characteristic patchy distributions (Rohrig and Ulrich 1991). Large clonal patches discriminated by unique multilocus genotypes, often occupying an area of 10×30 m or more, were detected among the root-sucker populations (Kitamura *et al.* 2001).

Future critical studies on adaptive mechanisms and demographic genetic analyses of these local populations of the American beech hopefully will shed light upon the origin of various genotypes and substructuring of local as well as patch populations.

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